There were so many victims of the Spanish flu epidemic that several makeshift hospitals—such as this gymnasium in the US—had to be created.

30 MILLION

THE NUMBER OF **PEOPLE** WHO **DIED** OF **SPANISH FLU** IN **SIX MONTHS** OF **1918**

CONFIRMING THE GENERAL THEORY OF RELATIVITY

In his theory of general relativity, Einstein had predicted that the strong gravitational field of the Sun would warp space-time. This distortion would deflect light coming from the stars, causing their apparent position to differ from their real position. Independent observations by Arthur Eddington and Andrew Crommelin confirmed this theory. During a solar eclipse, they accurately measured the positions of stars close to the Sun. They compared their results with the stars' apparent positions and noticed a slight difference between the two.

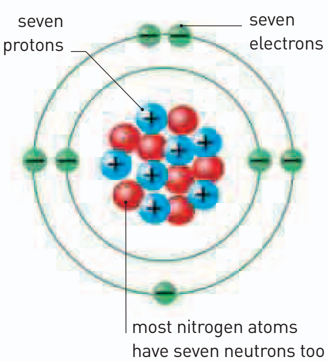
IN THE WAKE OF WORLD WAR I, a natural global disaster killed more people than the great plague of the 1300s. **Spanish flu** spread to all the continents, and medical science seemed powerless to stop it. Modern research suggests that it originated as a mutated virus and many people lacked the immunity to fight it.

In February 1918, French physicist Paul Langevin (1872–1946) demonstrated a system to detect the sound of **underwater submarines**—a technology that would have important implications in wartime. Called the **ASDICS** device (Aided Sonar Detection Integration and Classification System), it was the forerunner to **sonar** (see pp.292–93).

In Germany, electrical engineer Albert Scherbius (1878–1929) invented a **cipher machine**, which he called **Enigma**. It worked by a series of rotating wheels. In 1918, Scherbius offered it to the German Navy. They began using it in February 1926, and the German Army followed shortly afterward.

In 1917, New Zealand-born physicist Ernest Rutherford had found that firing alpha particles from radioactive elements at nitrogen atoms yielded small elementary particles. He realized that these were equivalent to **positive hydrogen ions** and later called them **protons**. It would be shown that they are responsible for the positive charge in the nucleus of atoms.

During a solar eclipse in 1919, British astronomer Arthur Eddington (1882–1944) and French astronomer Andrew Crommelin (1865–1939) observed that **rays of starlight were bent as they passed close to the Sun**. This proved **Einstein's general theory of relativity** (see pp.244–45).



Balanced charge
The number of protons in a non-ionized atom equals the number of electrons; this means that an atom, overall, has no charge. Neutrons are particles that have no charge.

Enigma cipher machine
These machines were used during World War II for decrypting and encrypting confidential messages.

January 8, 1918 American astronomer Harlow Shapley publishes his estimate of the **size and shape of the Milky Way** and deduces that it is not centered on Earth

January 1918 Spanish **flu pandemic** first identified in Kansas

February 1918 Paul Langevin demonstrates **ASDICS**—the forerunner to modern sonar systems

April 18, 1918 Arthur Scherbius offers his **Enigma cipher machine** to the German Navy

June 1918 British physicist Ernest Rutherford explains that **hydrogen ions** are **elementary particles**

May 29, 1919 Arthur Eddington and Andrew Crommelin test **Einstein's general theory of relativity**

June 1, 1919 American chemist Irving Langmuir describes the **arrangement of electrons** in atoms and molecules, and introduces the term **covalent bond**

November 30, 1919 Spanish flu pandemic is declared at an **end**

December 1, 1919 American chemist Phoebus Levens describes **nucleotide composition of DNA**